



ANKARA BİLİM
ÜNİVERSİTESİ

CENG 361 Web Design and Programming
Project Report

Online Skill-Based Quiz System

Student Name: İbrahim Türkel

Student ID: 220201047

Student Name: Kerem Çermikli

Student ID: 230201028

Student Name: Yiğithan Erdoğan

Student ID: 230201048

1. INTRODUCTION

This report outlines the development process, goals, and outcomes of our Online Skill-Based Quiz System. We undertook this project to create a reliable digital assessment tool that solves common issues found in traditional testing methods, and poor user engagement.

1.1 Purpose

Our main goal was to build a fully functional web platform where users can take subject-specific quizzes under controlled conditions, and administrators can easily manage the content.

Our motivation was to provide a fair, accessible, and secure testing environment that could theoretically be used for real classroom assessments.

1.2 Project Scope

Included Features (What we delivered):

- **Secure Access:** We built a registration and login system that protects user passwords using `password_hash` rather than storing them in plain text.
- **Quiz Engine:** Admins can create quizzes with specific time limits and subjects. The system supports both multiple-choice and short-answer questions.
- **Real-Time Testing:** When a user takes a quiz, a JavaScript timer counts down. If the time runs out, the system automatically submits their answers to ensure fairness.
- **Gamification:** To make it more engaging than a standard test, we added a Leaderboard to show top scores and a Badges system (like Expert or Ambitious) that users earn by completing quizzes.
- **History & Profiles:** Users can visit their profile to see every quiz they've taken, their score percentages, and the badges they have collected.

Out of Scope (Constraints):

- We did not develop a mobile app; the project is a responsive web application.

1.3 Innovative Aspect of the Project

What sets our project apart is the combination of Capacity Management and Gamification. Unlike static forms, our system is capacity-aware, meaning it can control quiz participation based on available resources.

Furthermore, by moving away from boring interface designs and adding competitive elements like leaderboards and badges, we turned a standard assessment tool into a more modern, engaging experience for students.

1.4 Global Goals for Sustainable Development (SDG Alignment)

1.4.1 Impacts on Society, Health & Safety, Economy, Sustainability, and the Environment

Socially, our platform democratizes education by allowing users to test their skills from anywhere, removing physical barriers to learning. Environmentally, moving assessments online significantly reduces paper waste. Economically, automating the grading process saves time and administrative costs compared to manual exam processing.

1.4.2 Project Formulation for Global Goals

Our project aligns closely with SDG 4: Quality Education. By building a scalable, open-source assessment tool, we are contributing to the digital infrastructure needed for modern education. It ensures that students can receive standardized feedback on their skills, regardless of their location.

2. LITERATURE REVIEW

Before starting, we looked at existing quiz platforms. Most simple solutions just calculate a score at the end without saving data, while professional platforms are often paid services. Research in web-based learning highlights the importance of usability and data persistence.

Our project fills the gap between these two extremes. We used a robust PHP backend and MySQL database to ensure that every attempt is recorded permanently. Unlike basic JavaScript quizzes that can be cheated, our system validates scores on the server side, ensuring a reliable assessment environment.

3. PROPOSED METHOD

We adopted a structured, layered approach to build this application, ensuring the code was clean and manageable.

Technologies Used:

- Backend: We used PHP for all the server-side logic, including calculating scores and managing sessions.
- Database: We designed a normalized MySQL database to store Users, Quizzes, Questions, and Attempts efficiently.
- Frontend: We used HTML and CSS with a custom Dark Mode theme to make it look modern. We utilized JavaScript specifically for the countdown timer to handle the client-side exam logic.

How it Works:

1. Database: We first designed tables like attempts and answers to link users to their quiz results.
2. Auth: We created a central auth.php file so we could easily protect pages like the Admin Panel from unauthorized users.
3. Quiz Logic: The take_quiz.php page loads the questions, and submit_quiz.php processes the results in a transaction to ensure no data is lost.

4. IMPLEMENTATION AND RESULTS

We successfully implemented all the core features we planned at the beginning of the semester.

4.1 Test Results

- **Functionality:** We tested the entire flow—registering a new user, logging in, taking a quiz, and checking the leaderboard. Everything worked as expected.
- **Data Integrity:** We verified that when a user finishes a quiz, their score and detailed answers are correctly saved in the database history.
- **Timer:** We confirmed that if a user tries to stay on the quiz page longer than the allowed time, the JavaScript timer forces the submission.

4.2 Comparative Results & Performance Evaluation

- **Reliability:** Compared to a simple HTML form, our system is much more reliable because it calculates scores on the server.
- **Performance:** The application is lightweight. By using optimized SQL queries and a simple CSS file, the pages load very quickly even with multiple questions.

5. TEAMWORK AND COMMUNICATION

5.1 Team Roles and Collaboration

To complete the project efficiently, we divided the tasks based on our strengths:

- **Backend Logic:** İbrahim Türkel & Yiğithan Erdoğan & Kerem Çermikli focused on the PHP coding and ensuring the login/quiz logic worked correctly.
- **Database Design:** Kerem Çermikli & İbrahim Türkel & Yiğithan Erdoğan handled the database schema and relationships between tables.
- **Frontend & Design:** Yiğithan Erdoğan & İbrahim Türkel & Kerem Çermikli worked on the UI and the JavaScript timer.
- **Reporting:** Kerem Çermikli & İbrahim Türkel & Yiğithan Erdoğan compiled the documentation and reports.

5.2 Communication and Reporting

Since we couldn't always work side-by-side, we relied on WhatsApp for daily updates and quick questions. We also held online meetings (via Teams) to discuss bigger technical decisions, like how to structure the database. This constant communication helped us resolve bugs quickly.

6. CONCLUSION

In conclusion, we successfully designed and built the Online Skill-Based Quiz System. We managed to go beyond a simple form-filling website by adding features like capacity management and gamification elements.

Limitations and Future Work:

- **Current Limitations:** Right now, the system only supports exact-match grading for text answers, which can be strict if a user makes a typo.
- **Future Improvements:** In the future, we would love to add real-time notifications for admins and perhaps an adaptive mode where questions get harder as the user answers correctly.



